Sound: pitch and amplitude

antocciphysics

UNIT 6 / DAY 8 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a drawing or explanation of a longitudinal wave to distinguish pitch from sound level.

SUCCESS CRITERIA - I can...

1. Describe compressions and rarefactions.
2. Connect pitch primarily with frequency.
3. Separate particle motion from sound propagation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Light moves from a transparent medium with n=**1.6** into air (n=**1**).
1. **Find critical angle** using sin θ_c=**1**/n.
2. **Describe incidence above it** in this ideal case.

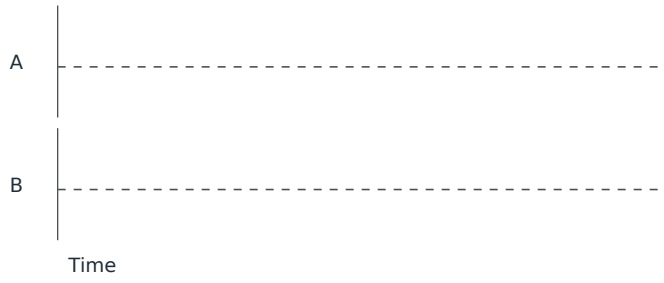
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: T = **1**/f; v = f λ.
Higher f generally means higher pitch. Amplitude affects intensity with the other parts of the setup kept the same, not frequency.

Worked reasoning: A tuning fork makes **440 cycles** each second, so its frequency is **440 Hz**. A louder version can have the same frequency but greater amplitude.

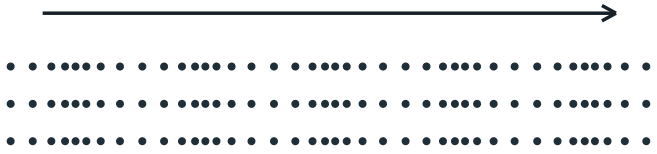
Your turn: Draw two displacement-time traces with equal periods and different amplitudes.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Mark compressions and rarefactions in the particle picture. Do air particles travel from speaker to listener with the disturbance?



Label dense and sparse regions.

Marzano check - circle: ① ② ③

04 Apply the physics

A tone generator plays **250 Hz** and **500 Hz** in the same air.
1. Use a sound speed of **340 m/s** to **calculate both wavelengths**.
2. Separately, double the pressure amplitude while frequency stays fixed. Use I ∝ A² to **compare intensity**.
3. **Label which feature changes** in each diagram.



Marzano check - circle: ① ② ③

Sound: pitch and amplitude

antocciphysics

Continue down each column.

05 Explain your reasoning

Turning up a speaker’s volume does not normally double a song’s pitch. **Explain which wave property mainly** changes.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Bodies initially at **340 K** and **360 K** exchange heat in an isolated system. **Explain what happens** to total energy and to the temperature-difference opportunity for work at equilibrium.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.

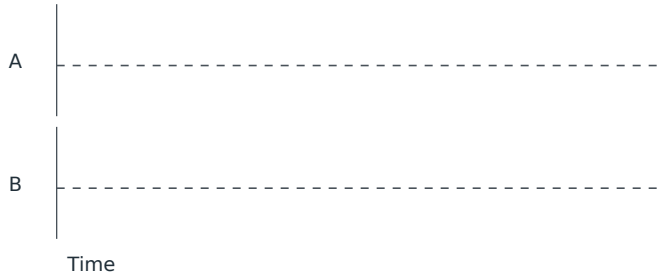
Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

A **500 Hz** tone lasts **0.10 s**. **Find number of cycles** and period.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two tones have frequencies **400** and **405 Hz**. Using beat frequency as their difference, find the beat rate. **Explain why beats** are not a new **405**-minus **400 Hz** pitch from either source alone.



Marzano check - circle: 1 2 3